

SmartShop Platform

E-Commerce System Architecture Project

From CRUD Operations to Scalable Transactional Design

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Strategic Project Objectives

Architectural Vision

To engineer a high-integrity e-commerce engine capable of managing full transactional lifecycles in a scalable and secure environment.

- **Transactional Flow:** Browse → Cart → Checkout → Order
- **Data Modeling:** Normalized relational structure (MLD)
- **Actor Separation:** Guest, Client, Admin, Server
- **Business Integrity:** Stock validation and atomic transactions
- **Security Layer:** Role-Based Access Control (RBAC)

Stakeholder Ecosystem

Guest

- Browse products
- View details
- Authentication access

Client

- Cart management
- Checkout
- Orders
- Reviews

Admin

- Product CRUD
- Order control
- Users management
- Stock control

Server

- Business logic
- Transactions
- Stock sync
- Validation

System Functionalities (Layered Architecture)

Frontend Layer

- Product browsing
- Product details view
- Cart interaction
- Checkout interface
- Order tracking

Business Logic Layer

- Cart validation
- Stock verification
- Order creation
- Price calculation

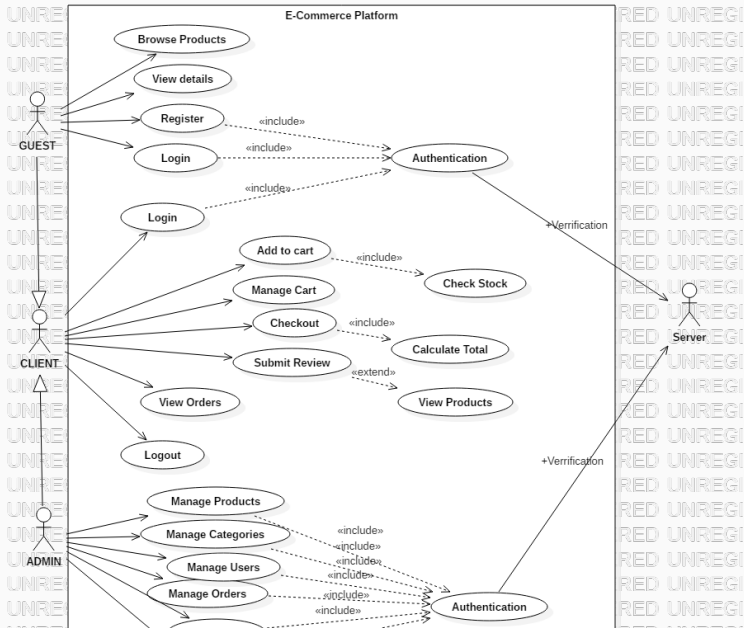
Backend Layer

- Product management (CRUD)
- Category management
- User management
- Order processing

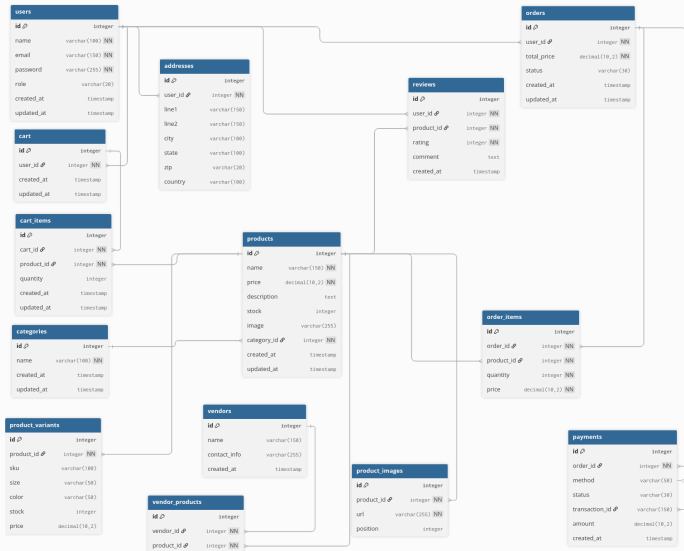
Security Layer

- Authentication (login/register)
- Role-based access
- Session handling
- Protected routes

Use Case Diagram



Conceptual Data Model (MCD)



Logical Data Model (MLD)

Core Tables

- **USER** (id, name, email, password, role)
- **PRODUCT** (id, name, price, stock, #category_id)
- **CATEGORY** (id, name)
- **CART** (id, #user_id)
- **CART_ITEM** (id, #cart_id, #product_id, quantity)
- **ORDER** (id, #user_id, total_price, status)
- **ORDER_ITEM** (id, #order_id, #product_id, quantity, price)

MLD Extension (Scalability Layer)

- **PRODUCT_IMAGE** (id, #product_id, url, position)
- **PRODUCT_VARIANT** (id, #product_id, size, color, stock, price)
- **REVIEW** (id, #user_id, #product_id, rating, comment)
- **PAYMENT** (id, #order_id, method, status, transaction_id, amount)
- **ADDRESS** (id, #user_id, city, country)
- **VENDOR** (id, name, contact_info)
- **VENDOR_PRODUCT** (id, #vendor_id, #product_id)

Business Logic Rules

- Prevent checkout if cart is empty
- Validate stock before order creation
- Deduct stock after purchase
- Prevent duplicate cart items (increment quantity)
- Ensure atomic transactions

- Clean scalable architecture
- Strong relational database design
- Clear separation of actors and system logic
- Ready for real-world deployment extension

Thank You

Questions & Discussion